

The Magnetic Fields of Pulsating Cepheid Variables

James A. Barron¹, Gregg A. Wade², Colin P. Folsom³ and Oleg Kochukhov⁴

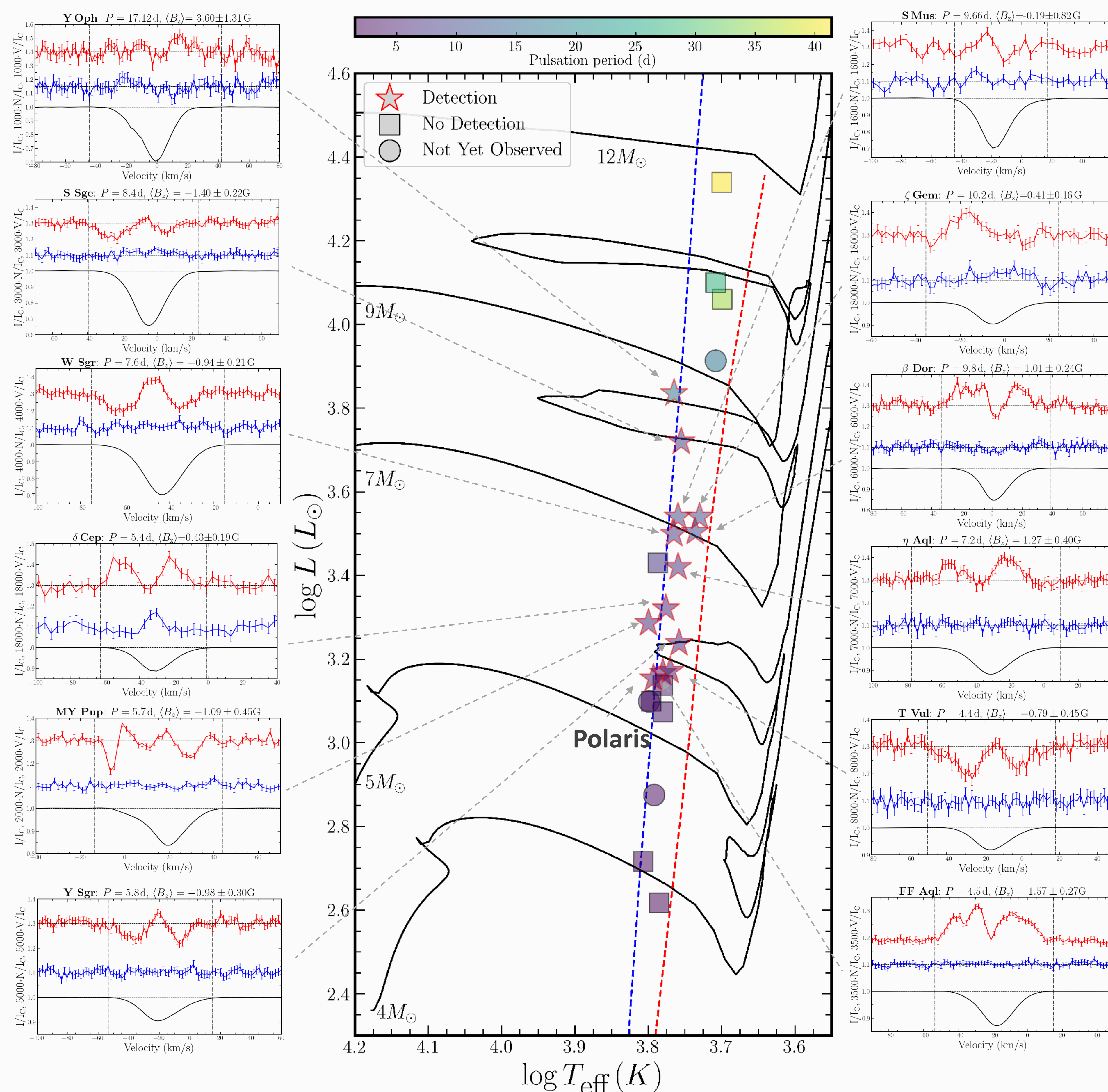
Background

- Classical Cepheids are essential tools for studying cosmology and stellar evolution due to their radial pulsations and period-luminosity relation (Leavitt Law).
- Little is known about the magnetic fields of Cepheids and their impact on Cepheid evolution and stellar properties.
- Previously η Aql was the only Cepheid to have a confirmed magnetic detection [2].

Survey

- We have identified a magnitude-limited sample ($V < 6$) of **twenty-five** Cepheids to perform a **first systematic magnetic study**.
- Twenty-one** targets have been observed to date with ESPaDOnS at the Canada-France-Hawaii Telescope and HARPSpol at the ESO 3.6m Telescope.
- Polarimetric spectra are obtained at high S/N (~ 4000 at 500 nm) and analyzed using Least Squares Deconvolution (LSD) to detect weak Stokes V magnetic signatures.

Diversity of Stokes V Magnetic Signatures

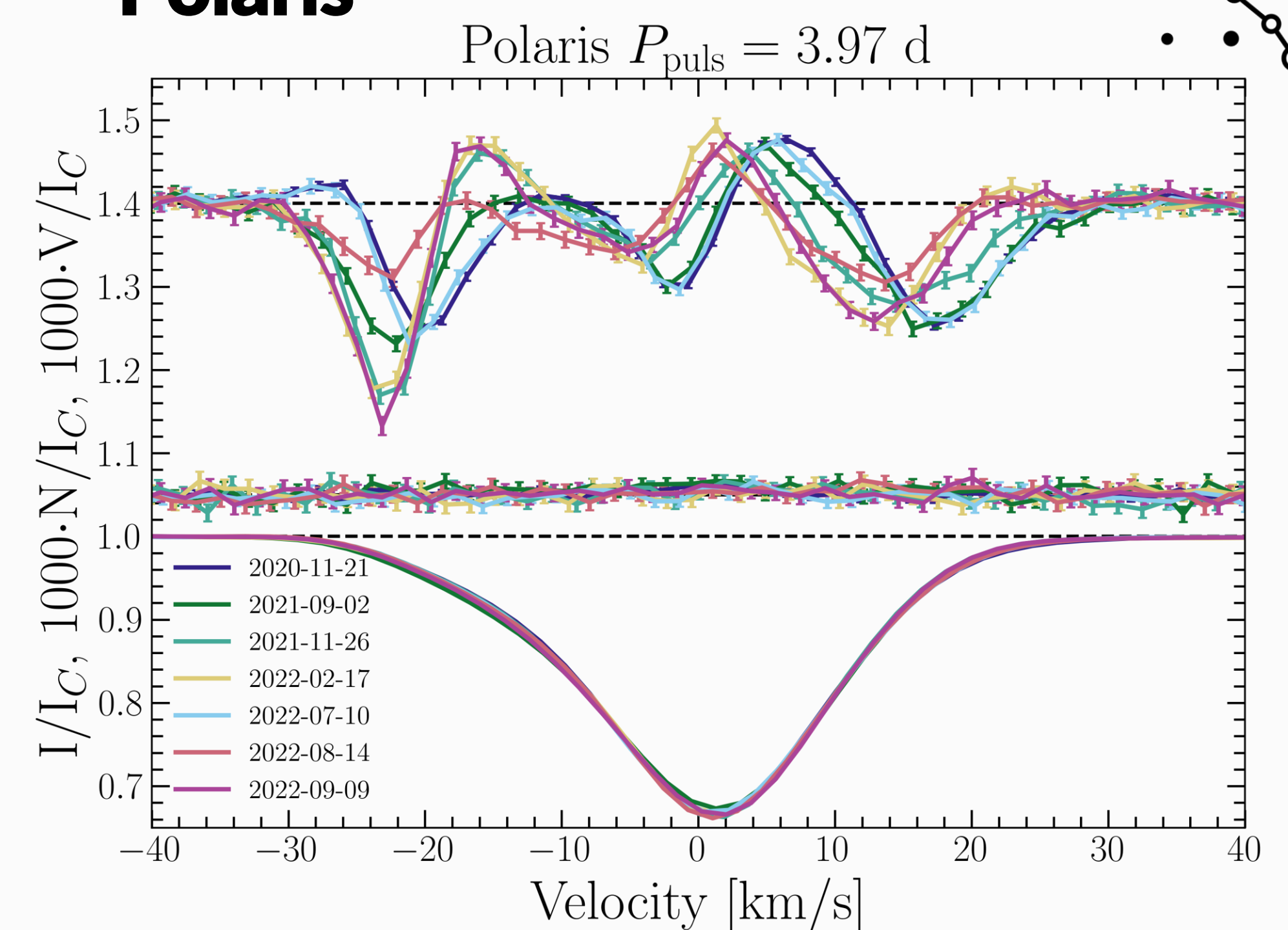


LSD Stokes I (black), V (red) and diagnostic null (blue) profiles for the Cepheid magnetic detections we have obtained to date. The Hertzsprung-Russell diagram illustrates that the sample is representative of a wide range of stellar parameters across the instability strip [3, 4, 5, 6].

Discussion

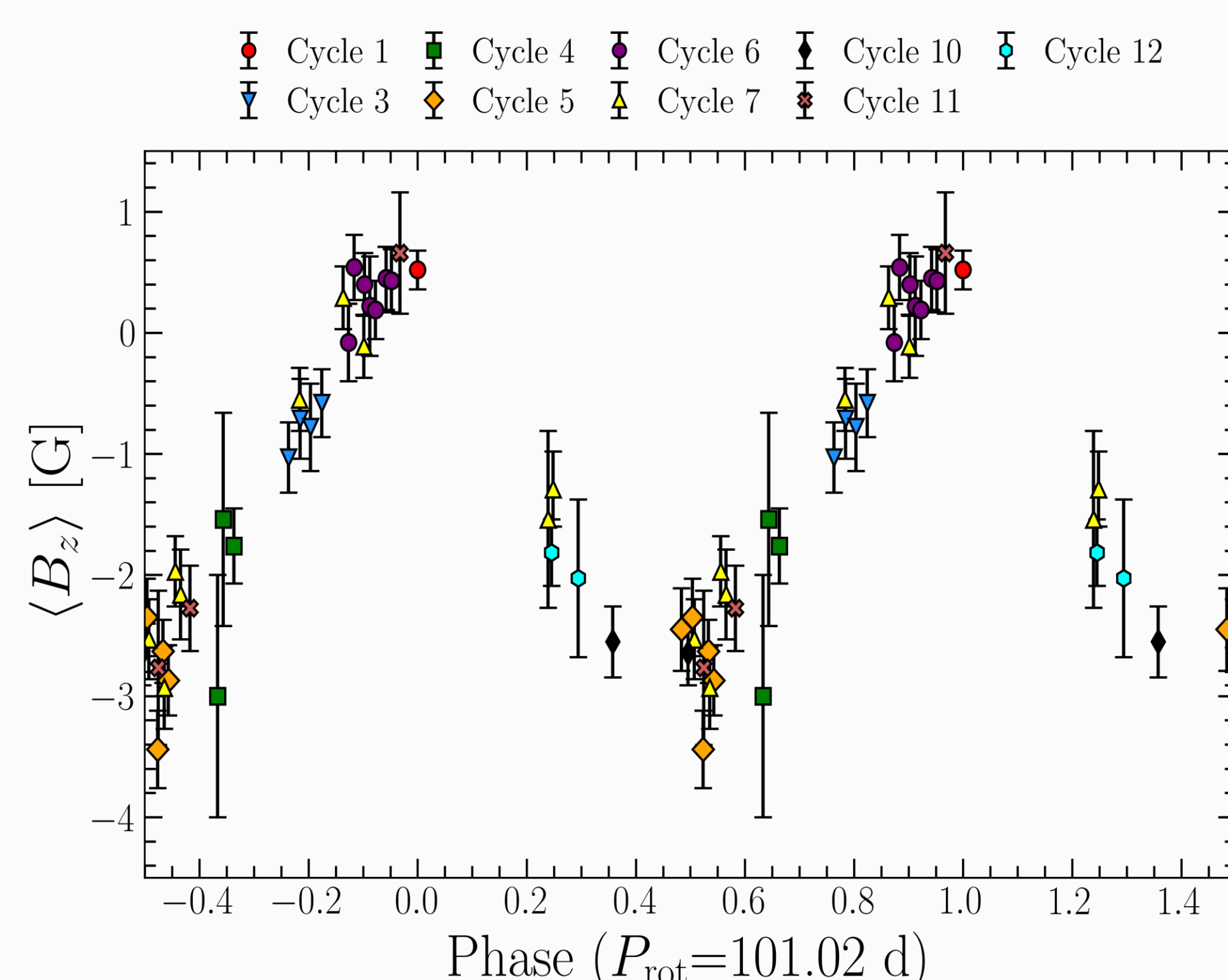
- We detect LSD Stokes V signatures in **thirteen** classical Cepheids, demonstrating that their magnetic fields are frequently detectable when observed with sufficient precision.
- Many targets show **peculiar unipolar positive or negative Stokes V lobes**, which are not predicted by standard Zeeman theory. We hypothesize that these features are due to the Zeeman effect modified by atmospheric velocity/magnetic field gradients and shocks [7, 8].
- These Stokes V profiles show **similarities** to those **detected in some Am stars** (e.g. Sirius A) [9]. Preliminary modelling that includes gradients can qualitatively reproduce these signatures [10].
- In contrast, the Stokes V profiles of Polaris and MY Pup appear more similar to those observed in cool non-pulsating supergiants [1]. We attribute this to their **low amplitude pulsations**.
- We are developing a hybrid modelling technique to combine Cepheid hydrostatic models and a polarized radiative transfer code. Modelling of Cepheid Stokes V signatures could provide a new and unique probe of Cepheid atmospheric dynamics.

Polaris



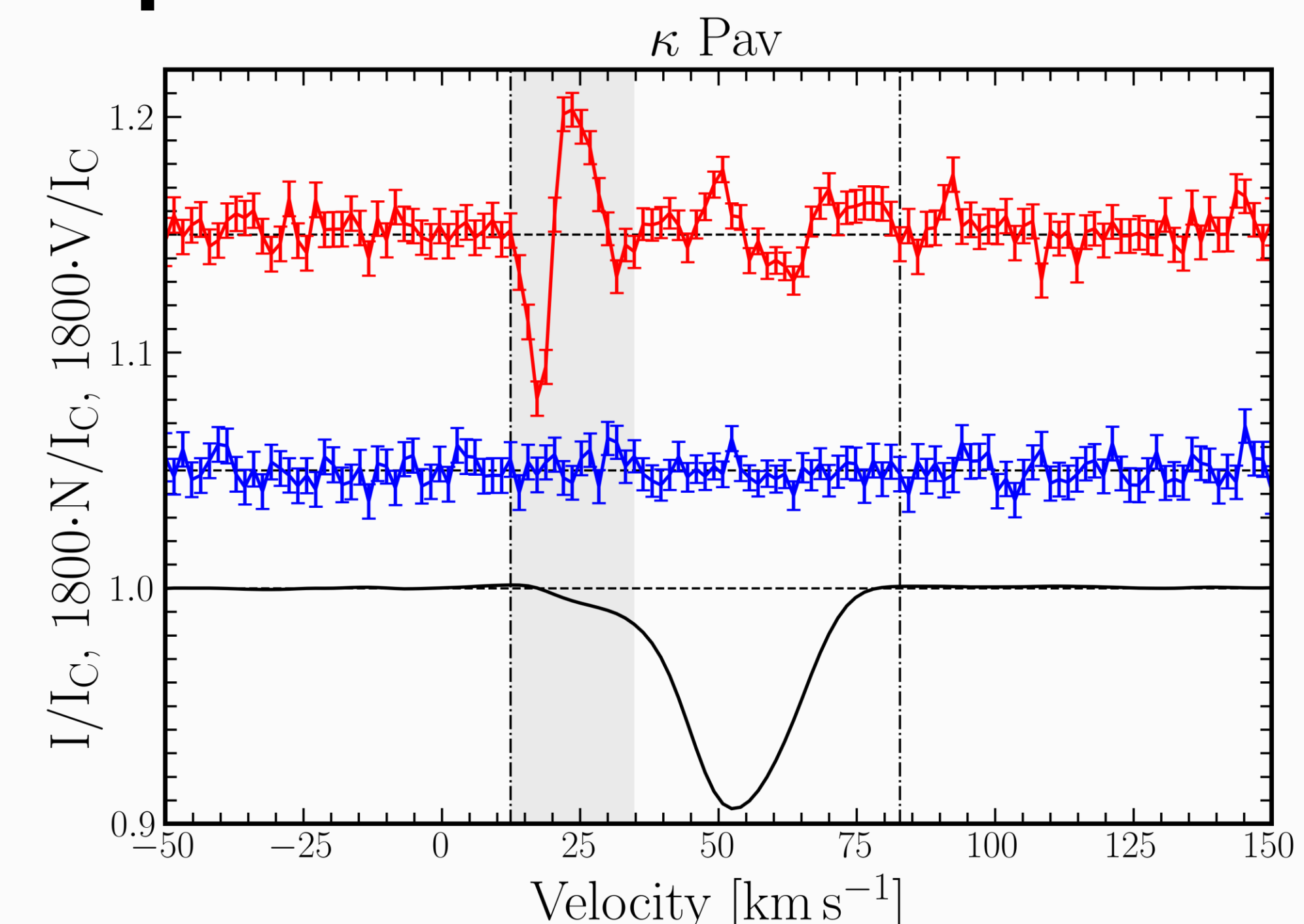
Representative Stokes V (top), null (middle) and Stokes I (bottom) profiles for Polaris obtained over seven ESPaDOnS runs spanning ~ 2 years (centered on 0 km/s).

The Stokes V signatures are detected at high S/N and indicate a **complex topology**.



Longitudinal magnetic field measurements of Polaris phased to a measured ~ 101 d rotation period. This is the **first direct measurement** of the **rotation period** of a classical Cepheid. The magnetic measurements have remained remarkably stable over 3 years of observation corresponding to 11 rotation cycles.

Impact of Shocks: κ Pav



LSD Stokes profiles of κ Pav, a **peculiar W Vir Type II Cepheid**. This is the **first magnetic detection** in a W Vir star. The observation was obtained just after the peak of the radial velocity curve as the infalling atmosphere begins to slow. The blue wing of the Stokes I profile shows a **shock feature** corresponding to an enhancement in the Stokes V profile.

Affiliations:

- Queen's University, Kingston, ON, Canada
- Royal Military College of Canada, Kingston, ON, Canada
- Tartu Observatory, University of Tartu, Tõravere, Tartumaa, Estonia
- Department of Physics and Astronomy, Uppsala University, Sweden

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